Sample Paper 9 — Answer Key (Level-2 Difficulty Step-Up)

Total: 40 marks (30 × 1 mark + 5 × 2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Cabbage** — Rose, Lily, and Jasmine are flowers; Cabbage is a vegetable.
2. **A) Sky** — A fish lives in water; a bird lives (flies) in the sky.
3. **A) WSPZI** — The rule is "+4 letters" (wrapping Z → D): S → W, O → S, L → P, V → Z, E → I.
4. **C) 45** — Two patterns are interleaved: the odd-position terms go 4, 8, 12, 16 (+4 each); the even-position terms go 60, 55, 50, ? (−5 each) — so the next even term is 45.
5. **A) Juhi, Hina, Gopal, Irfan, Farah** — Gopal is at position 3 (middle). Juhi is at the left end, position 1. Irfan is immediately right of Gopal, position 4. Hina sits between Juhi (1) and Gopal (3), so Hina is position 2. The only person left, Farah, takes position 5.
6. **A) 18 m North** — Both North legs add together (15 + 9 = 24 m North), and the 6 m South leg is subtracted: 24 − 6 = 18 m North.
7. **B) Saturday** — 26 − 9 = 17 days later; 17 = 2 weeks + 3 days, so the day of the week moves forward 3 days from Wednesday: Saturday.
8. **D) 46** — 14, 21, and 35 are all multiples of 7; 46 is not.
9. **C) 29** — Each term is the sum of the previous two: 11 + 18 = 29.
10. **B) 27** — Total students = (position from top) + (position from bottom) − 1 = 7 + 21 − 1 = 27.

Section 2 — Mathematical Reasoning

11. **A) 92468** — 67890 + 24578 = 92468.
12. **A) 47544** — 85000 − 37456 = 47544.
13. **A) 27402** — 4567 × 6 = 27402.
14. **A) 483** — 7245 ÷ 15 = 483 exactly.
15. **B) 14** — 42 = 2×3×7, 56 = 2^3×7. HCF = 2×7 = 14.
16. **C) 70** — 10 = 2×5, 14 = 2×7. LCM = 2×5×7 = 70.
17. **D) 7/11** — As decimals: 3/7 ≈ 0.429, 5/8 = 0.625, 4/9 ≈ 0.444, 7/11 ≈ 0.636. The greatest is 7/11.
18. **B) 8 km 150 m** — 5400 m + 2750 m = 8150 m = 8 km 150 m.
19. **C) 864 sq m** — Area = 36 × 24 = 864 sq m.
20. **C) 11 cm** — A regular nonagon has 9 equal sides: 99 ÷ 9 = 11 cm.

Section 3 — Everyday Mathematics

21. **C) ₹12,800** — 76800 ÷ 6 = 12800.
22. **C) 2,100** — Rate = 1800 ÷ 6 = 300 toys/hour; in 7 hours = 300 × 7 = 2,100.
23. **C) 22 min** — 2750 ÷ 125 = 22 minutes.
24. **C) 4 h 35 min** — Total time 8:00 a.m. to 12:50 p.m. = 4 h 50 min; subtracting the 15-minute stop gives 4 h 35 min of actual travel.
25. **C) 2 L 300 mL** — 5500 mL + 3700 mL = 9200 mL; 9200 ÷ 4 = 2300 mL per bottle.
26. **C) ₹160** — 8 × ₹22 = ₹176; ₹176 − ₹16 discount = ₹160.
27. **C) 900 L** — 1/8 × 1440 = 180 L used per day; over 3 days = 540 L used; 1440 − 540 = 900 L remaining.
28. **C) 50 kg** — 16400 g + 13600 g + 10800 g + 9200 g = 50000 g = 50 kg.
29. **D) 45** — 1080 ÷ 24 = 45 boxes.
30. **C) ₹280** — Sold for 70 × ₹44 = ₹3,080; profit = ₹3,080 − ₹2,800 = ₹280.

Section 4 — Achievers Section (2 marks each)

31. **C) 15** — Working backwards: (90 + 30) ÷ 8 = 120 ÷ 8 = 15.
32. **C) 30** — Let the first number be x. Then second = $2x$, third = $2x+18$. Sum: $x + 2x + (2x+18) = 168 \rightarrow 5x + 18 = 168 \rightarrow x = 30$.
33. **C) ₹2,000** — After spending 1/4 on books, 3/4 of the money remains. After spending 2/5 of that on stationery, 3/5 of 3/4 = 9/20 of the original remains. So 9/20 of the total = ₹900 → total = ₹2,000.
34. **C) 243** — Boys = (2 × 119) + 5 = 243. Check: 119 + 243 = 362. ✓
35. **D) 21** — Each term is the sum of the previous two: 8 + 13 = 21.

Scoring guide (out of 40, Level-2-difficulty paper): 28+ is a very strong result at this difficulty. 20–27 is solid. Below 20 — review *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.